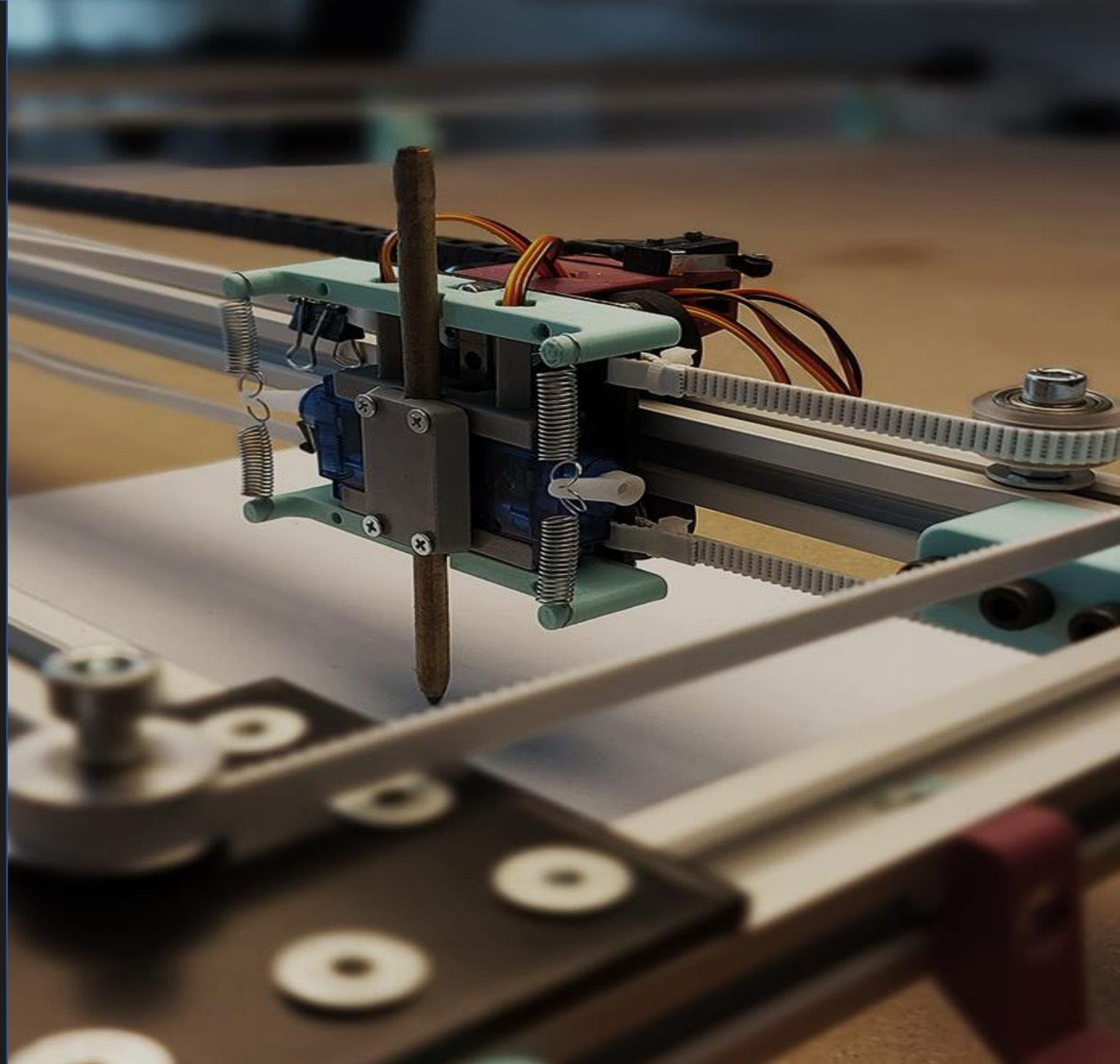
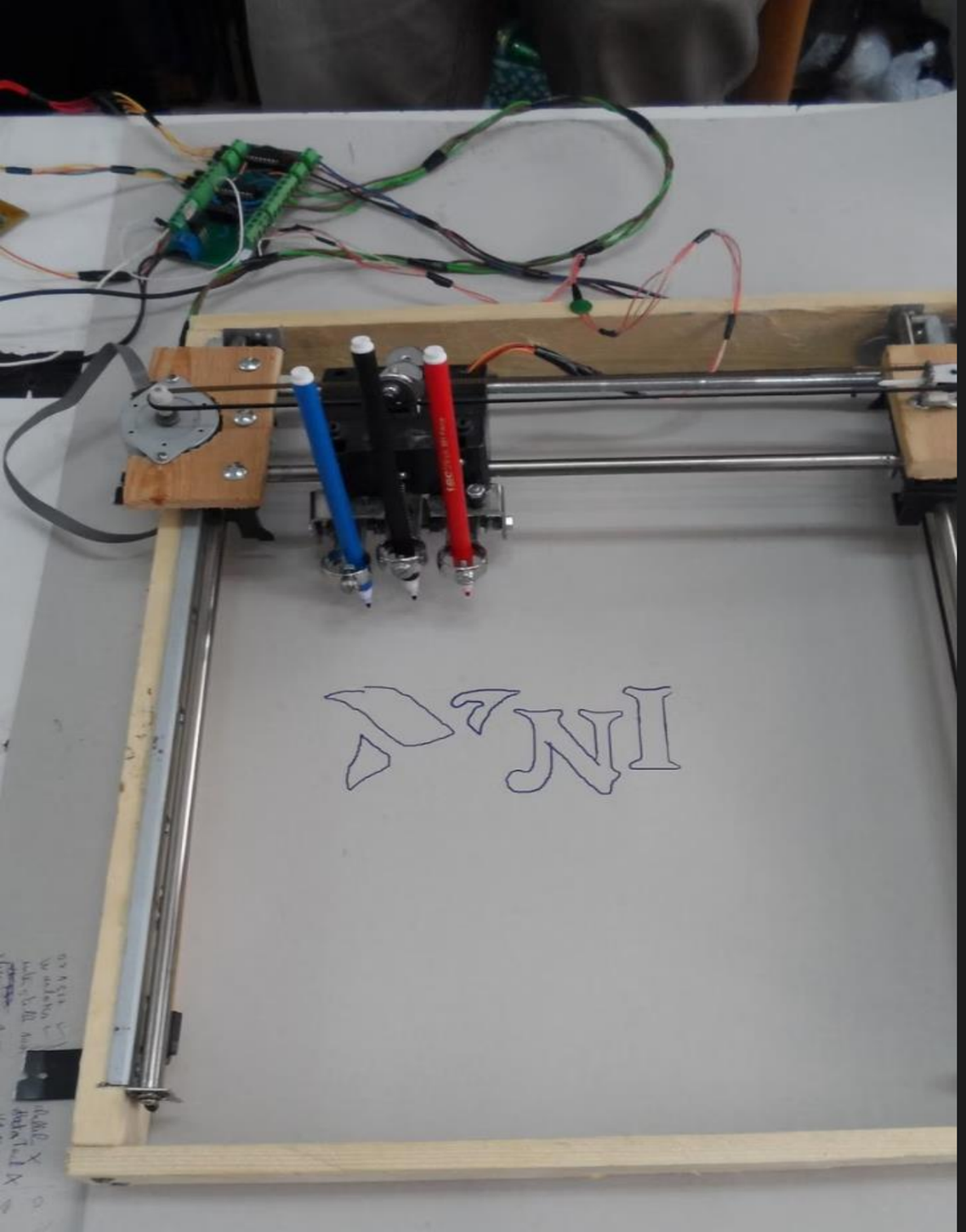
Drawi: A xy Plotter for Artistic Automation

Group Members:

Ami Sikder (JH-172-007)
Purba Ghosh (SK-172-008)
Mehedi Hasan Munna (AE-172-010)





Case Study:

Industrial Uses

- Automotive Manufacturing
- PCB Prototyping
- Laser Cutting

Advantages

- Precision
- Efficiency
- Versatility

Project Objective:

Develop Precise Plotter

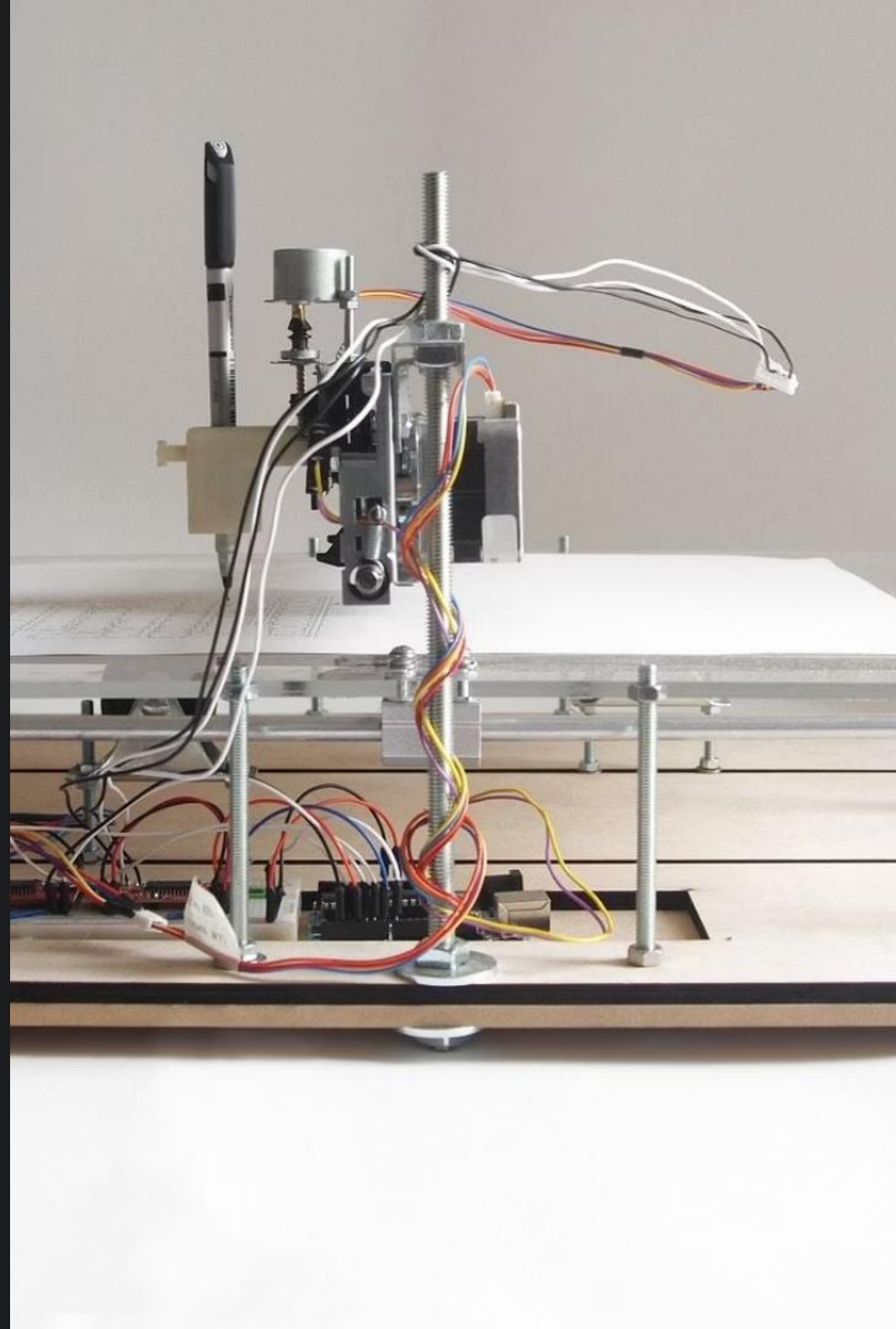
Create a controllable XY plotter system to accurately reproduce images

Implement Forward & Inverse Kinematics

Achieve accurate path planning and execution (±0.1mm)

Practical Application of Robotic Arm

The bot will demonstrate the application of the arm in 2D drawing context



Equipments:

Core Components

- Arduino Uno with CNC Shield
- NEMA 17 Stepper Motors (x2)
- Motor Drivers (A4988/DRV8825)

Structure & Connectivity

- Low-cost frame design
- GT2 belts, pulleys, rods
- 12V DC power supply
- Jumper wires, breadboard

Software & Tools

- GRBL firmware for Arduino
- Inkscape with G-code extension
- Universal G-code Sender
- Pen Control Servo

Methodology:

Mechanical Build
Construct XY plotter with belts and guides

Pen Control
Implement servo for pen lift/drop

G-Code Execution
Convert images to G-code; send via sender

Performance Evaluation
Assess drawing accuracy and repeatability

Electronics Integration
Connect CNC shield and GRBL firmware

Kinematic Conversion
Translate (X, Y) coordinates to motor steps

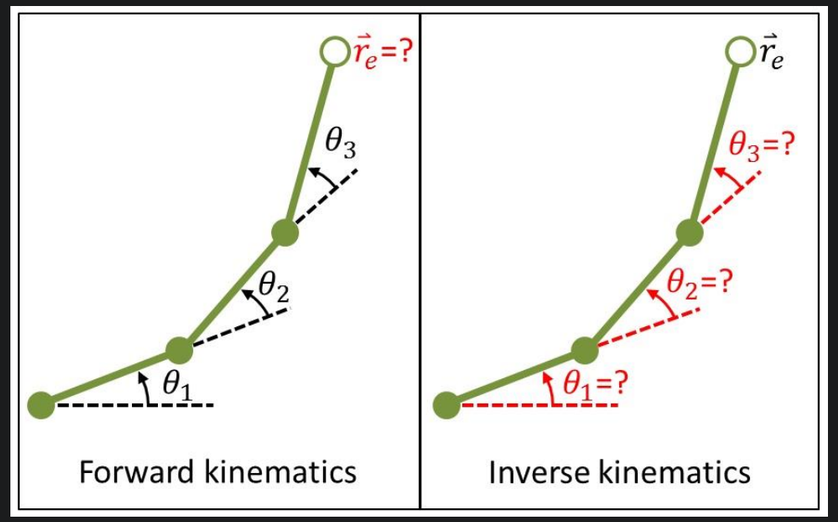
Calibration
Fine-tune stepper movement, pen offset, speed

Visualizing Kinematics: Forward and Inverse

Forward Kinematics

Joint angles determine end-effector position.

- Angles to position
- Direct calculation



Inverse Kinematics

Desired position calculates required joint angles.

- Position to angles
- Complex, iterative solutions

Conclusion:

Feasibility

Cost-effective solution for automated 2-D plotting

Educational impact

It's a practical platform for learning robotics

Future Applications

Potential for expanded use cases like laser engraving, PCB prototype and hand-written notes

THANK YOU

